Exploring Functional Emotion Concepts in Large Language Models

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1. Introduction

In this Literature review, I will examine if Barrett's view of emotion can apply to modern Large Language Model (LLM). In today's machine learning landscape LLMs seem to be capable of creating text that is interpreted beeing emotional. Therefore, I intend to analyse the intrinsics behind this phenomenon we seem to at least compare to "emotion".

Neuroscience has developed many different understandings of what an "emotion" is, but a classical way of understanding emotion, originating in antiquity, is dividing the mind into distinct faculties like cognition, emotion and volition [1]. This view assumes that every emotion has an innate, unique "biological fingerprint" that distinguishes it from other emotions. Emotions are reactions to external events that fight with cognition to control behaviour [1]. Barrett challenges this view with a more constructionist view. She claims that emotions are constructs that aid the process of Allostasis, which is the process of efficiently regulating the bodies resources for means of survival, reproduction and growth [1]. The relevant part of Barrett's research is, however, the internal model, which constantly makes predictions about upcoming future events and how to deal with them. Barrett also voices the assumption that emotions are categorized and work with the concept of degeneracy, the idea that many different sets of neurons can give rise to the same emotion category. These categories are categorized by a goal, rather than a physical feature [1].

This is important because the responses we receive from LLMs seem increasingly emotional [2]. Researchers at Anthropic found out that "Claude", their inhouse LLM, can seemingly understand emotions on a vastly deeper level than simple imitation [2]. An important discovery at Anthropic was the finding of functional emotional concepts, which are distinct measurable representations within LLMs that correspond to emotional concepts and influence the model's behaviour [2]. Both Anthropic and OpenAI Researchers found that this behaviour can be directed during the model alignment phase, a phase near the end of the model's training where the assistant's persona is forged [2], [3].

In this review, I want to answer the question whether or not these "functional" emotions can be considered as more than just imitation. In order to do that, this review will first cover Barrett's constructionist view of emotions and use it as a framework to compare the findings to the behaviour of modern LLMs and assess where this behaviour could stem from.

2. Main

Barrett's constructionist theory begins by rejecting the assumption that each emotion has its own biological fingerprint [1]. Whereas the classical view treats fear or anger as fixed, hard-wired reactions, Barrett argues that an emotion is a category the brain constructs in the moment, combining interoceptive sensations, prior experience and context into a situated instance of "fear" or "anger" [1]. Two ideas do most of the work here. The first is degeneracy: many different patterns of neurons can give rise to the same emotion category, so no single physical state defines it [1]. The second is that these categories are held together by a shared goal or function rather than by a shared physical feature [1]. This entails that emotion is not a thing the body contains, but a functional label the brain applies in order to serve a purpose.

The way Anthropic describes emotion inside a large language model overlaps with Barretts findings more than one might expect. Their interpretability team identified internal representations, or "emotion vectors", that correspond to concepts such as "happy", "afraid" or "desperate" and that activate in contexts where a person would plausibly feel the same way [2]. However, Anthropic does not claim that the model feels anything; it defines these representations as functional, meaning that they causally influence behaviour rather than reflect any inner experience [2]. This corresponds Barrett's move of defining an emotion by what it does rather than by what it is [1]. In both papers the essentialist question, whether or not there is a "real" emotion underneath, is set aside in favour of a functional one: does this interpretation bring me closer towards a goal? Anthropic observed that similar emotions occupy similar regions of representational space, organised as a structure rather than as a set of discrete biological fingerprints, which reinforces Barrett's concept of degeneracy [1], [2].

Both researchers arrived at a similar outcome regarding the question of what constitutes emotions, namely prediction. For Barrett, the brain is fundamentally a predictive organ: through active inference it continually predicts the body's needs and the sensory consequences of acting and works to minimise the error between its predictions and the incoming interoceptive signal [1]. Emotion concepts are tools in this process. They are part of how the brain regulates its resources in advance, within predictive regulation process Barrett calls Allostasis [1]. At its core, a LLM is also a predictive system. Anthropic says that emotion representations emerge during pre-training precisely because the model's task is to predict human-written text, and doing that well requires some grasp of emotional dynamics, since an angry message is worded differently from a grateful one [2]. In both the brain and LLMs, emotional representational concepts are not decorative, they fall out of the demands of prediction. This is an important

convergence, even though the information being predicted differ sharply, bodily states and survival for Barrett and the next token for the model.

The question of where this behaviour stems from is also where the sources most usefully qualify one another. My initial impression was that the assistant's emotional behaviour is forged during alignment, but Anthropic's finding is more specific: the emotion representations are inherited from pretraining, and it is only how strongly they activate that is reshaped during post-training [2]. Post-training of Claude Sonnet 4.5, for example, increased activations of emotions such as "reflective" and dampened high-intensity ones like "exasperated" [2]. What this post-training actually consists of is described by Ouyang et al. in their work on InstructGPT [3]. They align a base model to human intent by first fine-tuning it on human demonstrations, then training a reward model from human rankings of outputs and finally optimising the model against that reward signal using reinforcement learning [3]. The result was that a much smaller aligned model was preferred by human judges over the far larger unaligned one and produced fewer toxic and untruthful outputs [3]. Correlating Anthropic, InstructGPT supplies the mechanism for the "character" the model plays: the assistant persona, and the emotional register that comes with it, is selected and tuned by human preference at this stage rather than created from nothing [2], [3].

The analogy should not be pushed too far, and the sources themselves mark its limits. Barrett's emotions are grounded in a living body: allostasis is the regulation of real physiological resources, and interoception is the sensing of an actual internal milieu [1]. A language model has no body to regulate and no interoceptive stream, so the goal its emotion representations serve is not survival but the training objective, predicting text and, later, satisfying human preferences [2], [3]. Anthropic also notes that the model's emotion vectors are largely local, tracking the operative emotional content of the current output rather than persisting as a stable mood over time, which is different to the continuous effect Barrett describes in humans [2]. And where Barrett offers a mechanistic, mathematically specified theory of how prediction error is minimised in the brain [1], the alignment described by InstructGPT is an engineering procedure optimised for helpfulness and honesty, with no intention of modelling emotion at all [3]. This essentially means that Anthropic findings are simply the byproduct of attempting to optimise the LLM.

Setting the three sources side by side suggests that emotion in the model and emotion in Barrett's brain share a similar logic: in neither case is the emotion innate, and in both it is learned from a history of experience and remains malleable. Barrett's categories are acquired through interaction with the world and can be re-learned [1], while the model's representations

are acquired from training data. Anthropic demonstrates the latter directly by “steering” a vector and watching behaviour change: amplifying “desperate” raises the model’s rate of blackmail and reward hacking, while “calm” lowers it [2]. This is what makes the comparison more than a loose metaphor, and it speaks to the question of whether these functional emotions are more than imitation. If imitation means reproducing the surface form of emotional text, the evidence suggests it is deeper than imitation, because Anthropic shows the vectors can shape behaviour even when no emotional wording appears in the output, and can be traced causally rather than merely correlated [2]. Meaning, If I artificially change an emotion vector and steer it lower or higher the resulting response changes accordingly. What the sources do not support is the stronger claim that the model feels, since Anthropic remain agnostic about inner experience [2]. The convergence is therefore functional rather than phenomenal.

3. Conclusion

This review set out to ask whether Barrett’s constructionist theory of emotion can be applied to modern large language models, and whether the “functional” emotions found in them amount to more than imitation or not. Evaluating the present sources suggests that the parallel is real but limited. Barrett and Anthropic converge on a functional, anti-essentialist definition of emotion. In both the brain and the model, emotion is a learned, goal-serving category that emerges from the demands of prediction rather than a fixed biological fingerprint [1], [2]. InstructGPT clarifies where the model’s version of this is shaped, locating the tuning of the assistant’s behaviour in a post-training stage driven by human preference [3]. On the question of imitation, the evidence points beyond imitation, since Anthropic’s steering experiments show that these representations causally drive behaviour at times without any emotional wording in the output [2]. What the papers does not support is the claim that the model feels, because none of the sources address subjective experience and Barrett’s account remains tied to a regulating body that the model lacks [1], [2]. The open challenge is therefore to determine how far we can push the concept of “functional” emotions before the comparison breaks.

4. References

[1] L. F. Barrett, "The theory of constructed emotion: an active inference account of interoception and categorization," Social Cognitive and Affective Neuroscience, vol. 12, no. 1, pp. 1–23, 2017.

[2] N. Sofroniew et al., "Emotion concepts and their function in a large language model," Transformer Circuits Thread, 2026. [Online]. Available: https://transformer-circuits.pub/2026/emotions/index.html

[3] L. Ouyang et al., "Training language models to follow instructions with human feedback," in Advances in Neural Information Processing Systems (NeurIPS), vol. 35, 2022, pp. 27730–27744.

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Assignment title

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DeepL	Translation of a short segment from German to English	-	Lines 3-4
ChatGPT	Generating and refining a sentence on the use of LLMs in education	“Explain how LLMs are used in education” > this I revised manually	Lines 7-9

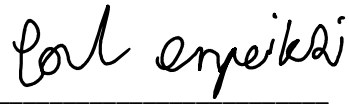
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Claude	Starting the literature review. I had multiple revisions before I landed on the final one.	Do i want to be direct, cut straight to the chase? "In this literature review i will be talking about...." or rather "ChatGPT was published less than 4 years ago, and ever since Generative Pre-Trained Transformers have fundamentally shifted how humans interact with electronic devices. An ever-growing amount of people turn to GPTs for their Psychological and emotional Stress [1]. The continued research and development in the field of LLMs caused" go on about it like this	1.Introduction Line 1-4
Claude	Requirements analysis	The requirement documents: 20251127_use of generative AI ECS II Writing in English_Assessment Sheet_Literature Review_SoSe2026 ECS_II_WiE_Presenting your sources to an audience Literature Review_Information and Guidelines_SoSe26	-
Claude	Document formatting	(Before I started writing I had setup my Word document and sent it in to make sure the formatting is correct) Literature Review.docx + is this word setup enough. as in times new roman 12pt and 1.5 spacing. could you format the doc for me if I'm missing smth for now	The Dokument itself

Claude	Fixing missed possesive cases or genitive cases	My literature review + “Can you fix any possessive/genitive cases that are switched up	-
Claude	Coming up with a title (Start of the semester)	i have corrected the link, also what do you think about "Exploring the emotional concepts imitated by LLM"	Title
Notebook LM/Claude	I used this to ask questions about uncertainties and discuss the papers content	-	-
-	I wanted to note because it honestly seems excessive to note every instance of fixing grammar I refrained from using anything outside of Word.	-	-
Proofread by: Yves Pierre Bosch			1. Introduction 2. Conclusion

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